JOÃO ALVIM

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EDUCATION

University of Minho — **October 2020 - October 2025**
Master's, Computer Engineering Formal Methods

Ghent University, Belgium — **October 2024 - March 2025**
Erasmus – Master's, Computer Engineering (1 semester) — *Belgium*

Escola Secundária Alberto Sampaio — **September 2017 - June 2020**
Secondary Education, Sciences

Conservatório Bomfim — **September 2017 - June 2020**
Secondary Education, Music – 8th Grade in Piano

WORK EXPERIENCE

INVISIBLE LAB — **January 2026 - Present**
Backend Software Engineer
Developed robust backend services in Golang backed by PostgreSQL. Designed external API integrations and delivered key features for Infocarro and a non-profit association platform.

DESY — **June 2024 - August 2024**
Summer Internship — *Hamburg, Germany*
Developed a monitoring dashboard with granafa for ASAPO – high performance distributed streaming platform for DESY a German research center operating particle accelerators.

2022 - 2025
Programming Tutoring to Undergraduates — *Braga*
Taught Haskell and C to 4 and 5 students, Natural Language Processing with Python to 1 student, and statistics with R.

Chess Club, University of Minho — **2022 - 2024**
Coach — *Braga*
Gave some weekly lessons where we analyzed textbook positions and also organized chess tournaments.

Tempos Brilhantes — **September 2023 - June 2024**
Elementary Chess Teacher — *Braga*
Biweekly extracurricular chess class to primary school kids.

Technician in High-Altitude Work — **1 week**

Monitor, Diver Lanhoso Adventure Park — **2 weeks**

PUBLICATIONS

- Online Abstract Debugging. SLATE 2026 – A debugging framework that allows abstract interpretation to be applied on-demand, gaining precision at specific points of program analysis before obtaining the full static results. It allows the user to unroll loops and explore intermediate states where one would normally lose precision. Out of the Loop no More

PROJECTS

pocket-cube-solver pocket-cube-solver
- Haskell solution to the Pocket Cube from Justin with.

Chess-Heatmaps Chess-Heatmaps
- Represent chess moves as heat maps in Python.

QForms QForms
- QuickForms is a self-hosted local form generation tool designed as an alternative to proprietary services like Google Forms. I've used Flask and my Python module is on PyPI.

vscode-ipma vscode-ipma
- VS Code extension that shows a popup with a beautifull IPMA's weather map.

cbirds cbirds

- Python module demonstrating combinator birds in point-free style.

shipslogs shipslogs

- Python tool that simplifies keeping records of screenshots taken during events such as talks, meetings or online classes.

Instagram Puzzles Generator Instagram Puzzles Generator

- Automates generation of chess puzzle images published on my university chess club's Instagram.

chessredditstories chessredditstories

- Automated content creation that generates reels combining chess gameplay with Reddit stories using Python.

datalist datalist

- Implementation of all functions in Haskell's Data.List module.

hmusik hmusik

- Generating chord progressions with Haskell.

TECHNICAL SKILLS

Languages: Portuguese (fluent), English (B2 exam)
Programming: Haskell, C, C++, Python, Java, TypeScript, JavaScript, LaTeX, SQL, Linux, VIM, WebAssembly, Go, PostgreSQL, Machine Learning

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AWARDS

- **2nd place:** SEI 2021 Hackathon (shipslogs)
- **1st place:** SEI 2025 Programming Competition

VOLUNTEER

Camerata Minhota **January 2023 - 2025**
Co-Founder and member of a Chamber Music group at University of Minho.

Coderdojo **January 2020 - January 2023**
Mentor

Recriativo, Núcleo de Estudantes de Engenharia Informática **2021 - 2022**
Collaborator

EXTRA-CURRICULAR

- Led a cat rescue from a 15 meter tree: Climber rescues cat trapped in 15 meter tree for six days
- Federated chess and climbing athlete
- Participated in the Bugsbyte 2023 and 2025 hackathons
- Attended IFL 2023 – Implementation and Application of Functional Languages conference
- De-Googled: self-host my own files, photos and calendar at home with Nextcloud, for privacy and freedom